

PS 160 — Physics II – Final Exam

Instructor name: _____ Section: _____

Date: Sat 07 Dec 2024

Student name _____ Student ID: _____

Score: /230

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- RELAX.
 - By starting this exam, you commit yourself to Embry-Riddle Aeronautical University's Code of Conduct.
 - You are allowed the instructor-provided equation sheet and a hand-held calculator. Aid from other sources (internet, other people, etc) is not allowed.
 - Look over all questions before you begin, and attempt the easier ones first. Place your answers in the boxes provided.
 - There will be no partial credit. To receive full credit you must SHOW your work. Your handwritten work MUST be submitted in a clear, concise manner.
 - You will be graded on your best 230 out of a total of 271 points.
 - All problems are worth 10 points unless otherwise noted.
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Formulas from Midterm 1 and Midterm 2

$$\rho = \frac{m}{V}$$

$$p = \frac{F_{\perp}}{A}$$

$$\frac{dp}{dy} = -\rho g$$

$$p(y) = p_0 - \rho g y$$

$$\rho A v = \mathcal{C}$$

$$p + \rho g y + \frac{\rho v^2}{2} = \mathcal{C}'$$

$$\frac{d^2 x}{dt^2} + \omega^2 x = 0$$

$$x(t) = A \cos(\omega t + \phi)$$

$$\omega = \sqrt{\frac{k}{m}}$$

$$\omega = \sqrt{\frac{g}{\ell}}$$

$$\omega = \sqrt{\frac{mgd}{I}}$$

$$f = \frac{1}{T} = \frac{\omega}{2\pi}$$

$$\omega = 2\pi f = \frac{2\pi}{T}$$

$$T = \frac{1}{f} = \frac{2\pi}{\omega}$$

$$y(x, t) = A \cos(kx - \omega t)$$

$$k = \frac{2\pi}{\lambda}$$

$$v = \lambda f = \frac{\omega}{k}$$

$$v = \sqrt{\frac{F}{\mu}}$$

$$v = \sqrt{\frac{Y}{\rho}}$$

$$v = \sqrt{\frac{B}{\rho}}$$

$$f_n = \frac{nv}{2L} \quad \left(\text{or } \frac{nv}{4L} \right)$$

$$\lambda_n = \frac{2L}{n} \quad \left(\text{or } \frac{4L}{n} \right)$$

$$P_{av} = \frac{1}{2} \sqrt{\mu F} \omega^2 A^2$$

$$I = \frac{1}{2} \sqrt{\rho B} \omega^2 A^2 \quad \left(= \frac{p_{max}^2}{2\sqrt{\rho B}} \right)$$

$$p_{max} = BkA$$

$$I = \frac{P}{4\pi r^2}$$

$$\beta = (10 \text{ dB}) \log_{10} \left(\frac{I}{I_0} \right)$$

$$I_0 = 10^{-12} \text{ W/m}^2$$

$$f_L = f_S \left(\frac{v + v_L}{v + v_S} \right)$$

$$f_{beat} = f_a - f_b$$

$$\Delta L = \alpha L_0 \Delta T$$

$$\Delta V = \beta V_0 \Delta T$$

$$\beta = 3\alpha$$

$$Q = mc\Delta T = nC\Delta T$$

$$Q = \pm mL$$

$$C_V = \frac{f}{2}R$$

$$C_P = C_V + R$$

$$\gamma = \frac{C_P}{C_V}$$

$$C_V = 3R \text{ (solid)}$$

$$H = kA \frac{\Delta T}{L}$$

$$H = A\sigma T^4$$

$$pV = NkT$$

$$pV = nRT$$

$$k N_A = R$$

$$\frac{1}{2}m \left(v^2\right)_{av} = \frac{3}{2}kT$$

$$v_{rms} = \sqrt{\left(v^2\right)_{av}}$$

$$f(v) \propto v^2 e^{-mv^2/2kT}$$

$$T_A = T_B = T_C$$

$$dU = \delta Q - \delta W$$

$$W = Q = Q_H + Q_C \text{ (for a cycle)}$$

$$\Delta S \geq 0$$

$$W = \int p dV$$

$$\Delta S = \int \frac{\delta Q}{T}$$

$$S = k \ln w$$

$$e = \frac{W}{Q_H}$$

$$K = \frac{Q_C}{-W}$$

$$K = \frac{-Q_H}{-W} \text{ (heat pump in heating mode)}$$

$$\Delta S = \frac{Q_H}{T_H} + \frac{Q_C}{T_C} = 0$$

Knowledge Questions

1. **[2 points]** Write the formulas for the laws of reflection and refraction (Snell's law).
2. **[2 points]** Write the formulas for the index of refraction and wavelength in a medium.
3. **[2 points]** Write the formulas for the critical angle (total internal reflection) and Brewster's angle.
4. **[3 points]** Write the formulas for the thin lens/mirror equation and the lateral magnification.
5. **[2 points]** Write the formulas for the lensmaker's equation and the small angle (paraxial) approximation.
6. **[2 points]** Write the formulas for Malus's law and the angular magnification of a magnifying glass.

7. [**3 points**] Write the formula for the phase difference between two rays.

8. [**2 points**] Write the formulas for *constructive* interference for a double slit and *constructive* interference for diffraction gratings.

9. [**1 point**] Write the formula for the intensity of two-source interference.

10. [**2 points**] Write the formulas for the path length difference for thin film reflection and the angular radius of the first dark ring (Rayleigh criterion).

Calculation Questions

11. A slab of ice (92% of the density of water) floats on a freshwater lake. What minimum volume must the slab have for a 45.0-kg woman to be able to stand on it without getting her feet wet?

$$V_{min} = \boxed{} \text{ m}^3$$

12. A horizontal pipe of radius 0.23 m carries water at 3.3 m/s and pressure of 233 kPa. If the pipe narrows to a radius of 0.12 m, calculate the pressure p in kPa in the narrow pipe. The density of water is 1000 kg/m³.

$$p = \boxed{} \text{ kPa}$$

13. A simple pendulum has length 0.67 m and mass 6.9 kg. If the local acceleration of gravity on this planet is 5.9 m/s², calculate the period of the motion.

$$T = \boxed{} \text{ s}$$

14. A simple harmonic oscillator consists of a block of an unknown mass in kilograms and spring with spring constant 776 N/m on a frictionless horizontal surface. The block is moved a distance 1.55 m from equilibrium and released from rest. If the block's maximum speed at the equilibrium point is 12.1 m/s, calculate the mass of the block in kg.

$m =$ kg

15. [20 points] Suppose a wave function is given by $y(x, t) = 5 \cos(3t + 0.5x - 2.1)$ where all values are in MKS units. Then ...

(a) the amplitude is m

(b) the angular frequency is rad/s

(c) the wavenumber is rad/m

(d) the phase shift is rad

(e) the frequency is Hz

(f) the period is s

(g) the wavelength is m

(h) the propagation speed of the wave is m/s

(i) the transverse **speed** of the wave at $(x = 0, t = 0)$ is m/s

(j) the maximum transverse **speed** of any point on the wave is m/s

16. The sound intensity level a distance 11 m from a pneumatic drill is measured to be 122 dB. Predict the sound intensity level from the same drill at a distance of 56 m.

$$\beta = \boxed{} \text{ dB}$$

17. A train approaches the station platform at 32 m/s while the engineer blows the 670-Hz horn. What frequency does a person on the platform hear? ($v_{\text{sound}} = 342 \text{ m/s}$)

$$f = \boxed{} \text{ Hz}$$

18. A pipe open at both ends has a length of 0.5 m. Calculate the frequency of the $n = 2$ harmonic. Take the speed of sound to be 331 m/s.

$$f = \boxed{} \text{ Hz}$$

19. A metal bar of length 4.6 m is heated from -4°C to 31°C . If its coefficient of *linear* expansion is $3.7 \times 10^{-5} \text{ K}^{-1}$, calculate the bar's change in length (in mm).

$$\Delta L = \boxed{} \text{ mm}$$

20. A 5.2-kg block of unknown material is in its solid phase at its melting point of 8°C . An experiment measures that 10^6 J must be absorbed by the block in order to transform all of it to its liquid phase at the same temperature. Calculate its latent heat of fusion.

$$L_f = \boxed{} \text{ kJ/kg}$$

21. Calculate the total translational kinetic energy of 10.1 moles of an ideal gas at a temperature of 27°C .

$$K_{tr} = \boxed{} \text{ kJ}$$

22. An ideal gas increases in temperature from 220°C to 450°C, and at the same time the volume decreases by 50%. If the initial pressure is 89 kPa, what is the final pressure?

$$p = \boxed{} \text{ kPa}$$

23. Three moles of helium gas (monatomic, $M = 4.00 \text{ g/mol}$) absorb 2400 J of heat energy at constant volume. If the original rms speed of the atoms is 900 m/s, what is the final rms speed of the atoms?

$$v_{rms} = \boxed{} \text{ m/s}$$

24. An ideal diatomic gas expands isothermally from an initial volume of 0.7 m³ to a final volume of 3.3 m³. If the initial pressure is 152 kPa, calculate the energy transferred by heat Q during this process.

$$Q = \boxed{} \text{ ______ (insert units here)}$$

25. A monatomic ideal gas expands adiabatically from 0.7 m^3 to 2.5 m^3 . If the initial pressure is 281 kPa, calculate the final pressure in (kPa).

$$p = \boxed{} \text{ kPa}$$

26. An engine with an efficiency 0.16 performs 1400 J of work during one cycle. How much energy does it exhaust to the cold reservoir in one cycle?

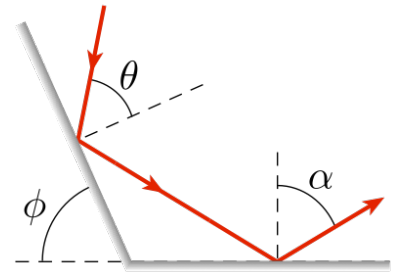
$$\text{(insert variable here) } _____ = \boxed{} _____ \text{ (insert units here)}$$

27. Suppose 11 kg of water at 0°C freezes to ice at 0°C . Calculate the entropy change of this process (in kJ/K). Carefully consider the sign of ΔS . (The latent heat of fusion of water is $3.35 \times 10^5 \text{ J/kg}$.)

$$\Delta S = \boxed{} \text{ kJ/K}$$

28. The figure shows two plane mirrors that meet at a corner. A ray of light is incident on the first mirror with an angle of incidence equal to θ . Calculate the angle α between the outgoing ray and the second mirror's normal if $\phi = 70^\circ$ and $\theta = 50^\circ$.

$\alpha =$ degrees



29. A ray of light traveling in air ($n_{air} = 1$) is incident at an unknown angle (with respect to the normal) on the surface of a fluid with index of refraction 1.6. If the angle of refraction is 20° , calculate the angle of incidence.

$\theta_{incidence} =$ degrees

30. **[20 points]** An object is located 18 cm in front of a concave spherical mirror with radius of curvature 16 cm. Determine the following:

(a) the image position, $s' =$ cm

(b) the lateral magnification, $m =$

(c) the image is (real/virtual)

(d) the image is (upright/inverted)

31. Light reflecting off a bubble in the shape of a spherical shell exhibits enhancement in reflected light at a wavelength of 471 nm. Calculate the bubble's minimum thickness (in nm) if its index of refraction is 1.5.

$$t = \boxed{} \text{ nm}$$

32. Monochromatic, coherent light of wavelength 418 nm is incident on two slits separated by 0.26 mm. If the screen is 2.1 m away, calculate the full width (in mm) of the central maximum on the screen.

$$\Delta y = \boxed{} \text{ mm}$$

33. Suppose a space telescope orbiting Mars has an aperture diameter of 4.5 cm. What minimum width crater (in meters) can it resolve in light of wavelength 557 nm if it is in an orbit 883 km from the planet's surface? Disregard atmospheric effects.

$$d_{min} = \boxed{} \text{ m}$$